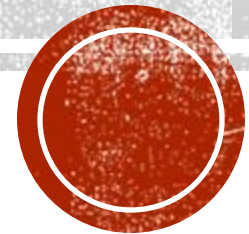


CLOUD COMPUTING

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SYLLABUS

1. Introduction to Cloud Computing
2. Cloud Computing Technologies
3. Networking in Cloud
4. Storage Solutions in the Cloud
5. Cloud Security and Compliance
6. Infrastructure as Code
7. Virtualization and Containerization
8. DevOps and Cloud



GRADING

- Attendance 10%
- Lab 20%
- Mid-term/Case studies 20%
- Project 20%
- Final exam 30%



WEEK 1: INTRODUCTION TO CLOUD COMPUTING

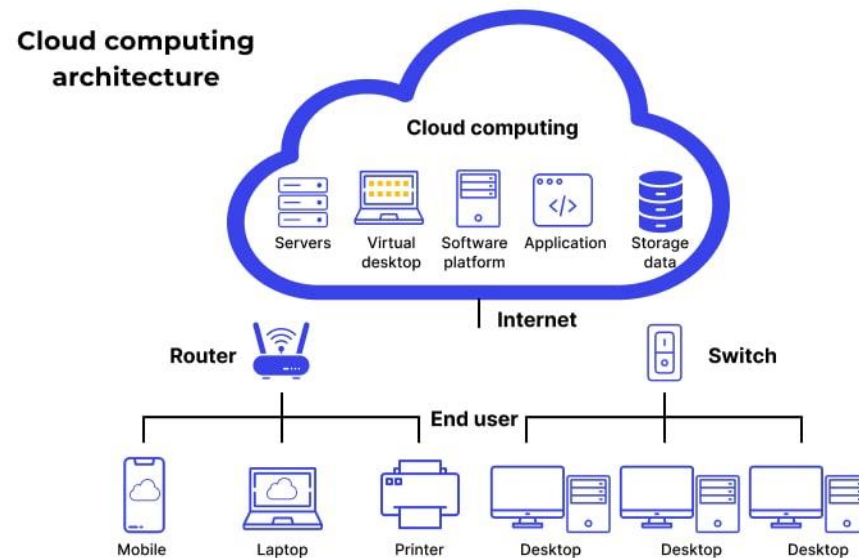
- **Learning Objectives:**

- Understand the definition and evolution of cloud computing.
- Identify key characteristics and benefits of cloud computing.
- Differentiate between various cloud deployment models: public, private, and hybrid.
- Differentiate between various cloud service models: IaaS, PaaS and SaaS.



WHAT IS CLOUD COMPUTING

- Cloud computing is the delivery of computing resources—such as servers, storage, databases, networking, software, and analytics—over the internet.
- It allows users to access and use them on demand often with **pay-as-you-go** pricing without needing to manage the underlying infrastructure.



REAL-WORLD ANALOGY

- "Think of cloud computing like electricity.
- You don't build your own power plant to get electricity—you just plug into the grid and pay for what you use.
- The same thing happens in cloud computing: you access computing resources from someone else's infrastructure and only pay for what you use."



HISTORICAL CONTEXT 1/2

- 1960s: Idea of computing as a utility
 - **John McCarthy**, a computer scientist, proposed that "**computation may someday be organized as a public utility**" — just like electricity or water.
 - At this time, **mainframe computers** were shared using **time-sharing** — a system where multiple users could access one powerful computer.
 - *This planted the seed of cloud computing: shared access to computing power.*
- 1990s: Internet + virtualization rise
 - The internet became more widespread.
 - **Virtualization technology** (e.g., VMware) allowed a single server to run multiple operating systems — key to cloud efficiency.
 - *The groundwork for infrastructure sharing and hosting was laid.*
- 1999: Salesforce launches cloud CRM
 - This was one of the first successful **Software as a Service (SaaS)** offerings.
 - *You could use business software online — no installation needed.*



HISTORICAL CONTEXT 2/2

- 2006: Amazon launches AWS
 - For the first time, developers and businesses could rent servers and storage by the hour.
 - *This marked the true beginning of cloud computing as a scalable, pay-as-you-go service.*
- 2010s: Cloud adoption accelerates
 - **Microsoft Azure** and **Google Cloud Platform** entered the scene.
 - Enterprises started moving from data centers to the cloud.
 - *Cloud became the default way to launch and grow digital services.*
- Today: Cloud powers AI, IoT, business, and more
 - Cloud providers now offer advanced services like: AI, ML, Big Data Analytics, IOT
 - Most companies use **multi-cloud** or **hybrid-cloud** strategies.
 - *Cloud is now essential for modern digital infrastructure.*



BENEFITS OF CLOUD COMPUTING 1/3

- **Cost saving**
 - No need to buy hardware or set up data centers.
 - Pay-as-you-go: You only pay for what you use (compute, storage, bandwidth).
 - Reduces costs on maintenance, IT staff, and electricity.
- **Scalability and Flexibility**
 - Instantly scale resources up or down based on demand.
 - Handle traffic spikes without over-provisioning.
 - Useful for seasonal businesses, app launches, and growing startups.
- **Global Accessibility**
 - Access your data, applications, and services from anywhere with internet.
 - Supports remote work and global teams.
 - Improves collaboration and productivity.



BENEFITS OF CLOUD COMPUTING 2/3

- **Faster Deployment and Time to Market**
 - Launch applications or infrastructure in minutes instead of weeks.
 - Speeds up development, testing, and deployment cycles.
- **Security and Compliance**
 - Cloud providers offer built-in security features, encryption, and identity management.
 - They often comply with global security standards (ISO, GDPR, HIPAA, etc.).
 - Regular backups and disaster recovery options are included.
- **Access to Advanced Technologies**
 - Easily use AI/ML, big data analytics, IoT, and blockchain services.
 - You don't need to build or manage the complex infrastructure yourself.



BENEFITS OF CLOUD COMPUTING 3/3

- Automatic Updates & Maintenance
 - No manual patching or upgrading servers.
 - Providers ensure that your systems are always up to date and running smoothly.
- Disaster Recovery and Business Continuity
 - Data is backed up and can be recovered across multiple data centers.
 - Ensures that your services stay available even if a server or region fails.



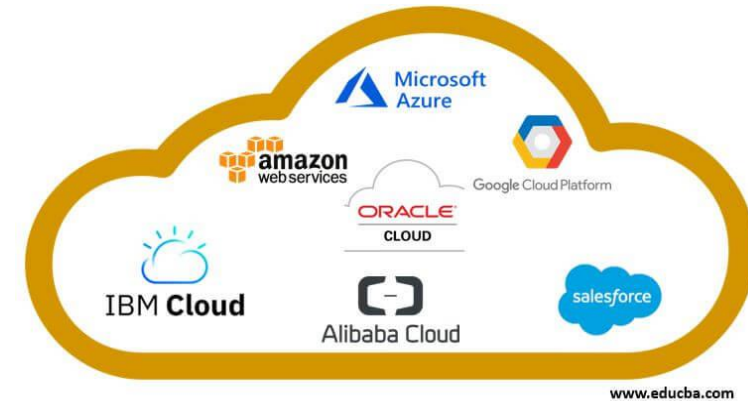
KEY CONSIDERATION

- Before you jump into the cloud, here are some things to consider:
 - **Security & Privacy:** Where is your data going?
 - **Cost Management:** Are you optimizing usage?
 - **Vendor Lock-in:** Can you switch providers easily?
 - **Compliance:** Are you following local regulations?
- Always ask: Is your organization **cloud-ready**?



CLOUD DEPLOYMENT MODELS 1/2

- **Public Cloud** services are shared among multiple organizations and is hosted by a third-party provider.
- **Private Cloud** exclusively used by a single organization. It can be hosted on-premises or in a dedicated environment, providing greater control over resources, security, and compliance.
- **Hybrid Cloud** combines both, allowing data and applications to be shared across them, optimizing both cost and security.
- **Community Cloud** is a cloud infrastructure shared by several organizations with common interests, such as security, compliance, or mission goals. It's like a private cloud, but instead of being used by one organization, it's shared by a specific group with common goals.

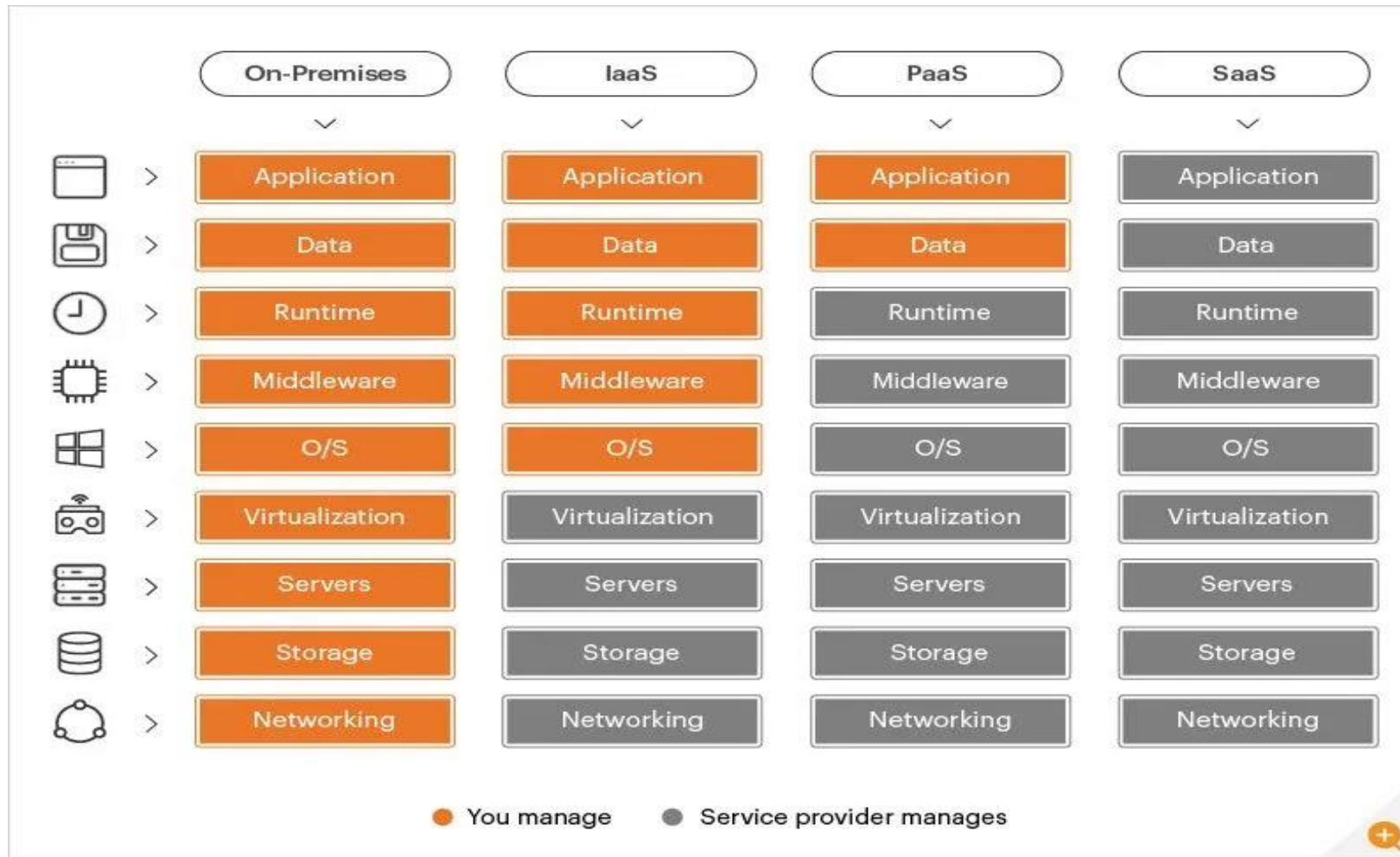


CLOUD DEPLOYMENT MODELS 2/2

Feature	Public Cloud	Private Cloud	Hybrid Cloud	Community Cloud
Ownership	Third-party	Single org	Mixed	Shared orgs
Cost	Low (shared)	High (exclusive)	Medium	Shared
Security	Standard	High	Flexible	High (group-based)
Scalability	High	Medium	High	Medium
Control	Low	Full	Mixed	Shared



CLOUD SERVICE MODEL



- **IaaS:** Infrastructure as a Service
- **PaaS:** Platform as a Service
- **SaaS:** Software as a Service



INFRASTRUCTURE AS A SERVICE

- IaaS contains the basic building blocks for cloud IT and typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space.
- **You manage:** OS, middleware, runtime, data, and apps
- **Cloud provider manages:** Physical hardware, networking, storage



PLATFORM AS A SERVICE

- PaaS vendors remove the need for organizations to manage the underlying infrastructure (usually hardware and operating systems), and this integration allows you to focus on the deployment and management of your applications.
- This helps you be more efficient, as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.
- **You manage:** Your applications and data
- **Cloud provider manages:** Servers, OS, middleware, runtime



SOFTWARE AS A SERVICE

- SaaS vendors provide you with software applications that are run and managed by the vendor. In most cases, people referring to SaaS are referring to third-party end-user applications.
- With a SaaS offering you do not have to worry about how the service is maintained or how the underlying infrastructure is managed; you only need think about how you will use that particular piece of software.
- A common example of a SaaS application is web-based email where you can send and receive email without having to manage feature additions or maintain the servers and operating systems that the email program is running on.
- **You manage:** Just your user data (or nothing at all!)
- **Cloud provider manages:** Everything else



FACTORS INFLUENCING CLOUD MODEL CHOICES

- **Cost considerations:** Budget constraints can heavily influence cloud choices.
- **Compliance and regulatory requirements:** Industries like finance and healthcare often have strict data regulations.
- **Security needs:** Some organizations prefer private clouds for sensitive data due to enhanced security controls.



Q & A

